ModelIt!

Teacher's Guide

Building for the Big One: Earthquake Engineering Lab

How Engineers Design Buildings That Bend Instead of Break

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-ESS3-2, 4-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-ESS3-2, 4-ETS1-2

Generate and compare multiple solutions to reduce the impacts of natural Earth processes on humans. Generate and compare multiple solutions to a problem based on how well they meet the criteria and constraints of the design problem.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Designing Solutions** — Students develop a five-component model of earthquake impact on buildings and use model evidence to design structures that minimize damage and maximize occupant safety.
Disciplinary Core Idea	**ESS3.B Natural Hazards / ETS1.B Developing Possible Solutions** — Natural hazards like earthquakes cannot be prevented, but their impacts can be reduced through engineering design. Testing and comparing multiple solutions helps identify the most effective approach.
Crosscutting Concept	**Cause and Effect** — Students investigate the causal chain from earthquake magnitude to shaking force to structural damage to occupant safety, and how engineering interventions (flexibility) can interrupt this chain.

CAST Testing Alignment

MULTIPLE CHOICE:

Two buildings are next to each other during a magnitude 7.0 earthquake. Building A is made of rigid unreinforced brick. Building B has a flexible steel frame with shock absorbers. Which building is most likely to survive? Why?

CONSTRUCTED RESPONSE:

Using what you know about earthquakes and engineering, explain why a building that can SWAY during an earthquake is actually SAFER than one that stays perfectly still. Use the words 'flexibility,' 'shaking force,' and 'structural damage' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Earthquakes are one of nature's most destructive forces, but they do not have to be deadly. The difference between a disaster with thousands of casualties and one where everyone walks away is often the ENGINEERING of the buildings. Understanding how buildings interact with earthquake forces is the key to saving lives.

How Earthquakes Work

Earthquakes happen when tectonic plates suddenly slip past each other along a fault line, releasing enormous amounts of stored energy as seismic waves. These waves travel through the ground and shake everything on the surface. Earthquake magnitude is measured on the Richter scale — each whole number increase represents about 32 times more energy. A magnitude 7 earthquake releases about 32 times more energy than a magnitude 6, and about 1,000 times more than a magnitude 5.

Rigid vs. Flexible: Why Buildings Fail

When the ground shakes during an earthquake, the bottom of a building moves with the ground but the top tries to stay still (due to inertia). This creates enormous stress in the structure. Rigid buildings — made of unreinforced brick, concrete blocks, or adobe — cannot flex to absorb this stress. Like a dry twig, they snap. The connections between walls and floors fail, and the building pancakes. Flexible buildings — with steel frames, reinforced concrete, and engineered connections — bend and sway with the motion, distributing the stress throughout the structure instead of concentrating it at weak points.

Engineering Solutions That Save Lives

Modern earthquake engineering uses several strategies. Base isolation places the building on rubber and steel pads that let the foundation slide horizontally while the building stays relatively still — like a book sliding on a table. Tuned mass dampers are heavy pendulums near the top of skyscrapers that swing in the opposite direction of the earthquake, counteracting the sway. Cross-bracing uses diagonal steel beams that resist lateral forces. Energy dissipation devices are essentially shock absorbers built into the building that convert shaking energy into heat.

The Cost of Not Building Better

The difference between earthquake engineering and no engineering is measured in lives. The 2010 Haiti earthquake (magnitude 7.0) killed over 200,000 people because most buildings were unreinforced concrete. The 2011 Japan earthquake (magnitude 9.0 — over 900 times stronger) killed about 20,000 people, mostly from the tsunami, because Japanese buildings are engineered for earthquakes. Japan experienced a much stronger earthquake but far fewer building collapses because they invested in engineering.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Earthquake Magnitude and Building Flexibility as external components (one natural, one engineered) and Shaking Force, Structural Damage, and Occupant Safety as internal components that result from the interaction.

E **Establish Relationships**

Students determine that Earthquake Magnitude positively affects Shaking Force, Building Flexibility negatively affects Structural Damage (flexible buildings resist damage), Shaking Force positively affects Structural Damage, Structural Damage negatively affects Occupant Safety, and Building Flexibility positively affects Occupant Safety.

V **Visualize & Model**

Students build a five-component model showing how earthquake force travels through a building and how engineering flexibility interrupts the damage chain to protect occupants.

E **Evaluate Outcomes**

Students run four scenarios comparing rigid vs. flexible buildings in small and large earthquakes, discovering that flexibility makes the biggest difference in large earthquakes.

R **Revise & Extend**

Students add foundation type or building height to explore base isolation and why taller buildings need special engineering to handle amplified swaying at the top.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at these two buildings. Same earthquake. Same street. One collapsed, the other is fine. What is the difference?"*

Watch For

Show before/after earthquake photos of collapsed vs. surviving buildings. Ask: What do you think the surviving building had that the other did not?

Slide 2: Learning Objectives

3 min

Say: *"Today you are going to become earthquake engineers. You will figure out WHY some buildings survive and design one that can handle the big one."*

Watch For

Have students read objectives aloud. Pre-teach 'flexibility' vs. 'rigidity' by bending a pipe cleaner vs. trying to bend a pencil.

Slide 3: Big Question

3 min

Say: *"If two buildings are right next to each other and feel the SAME earthquake, why would one collapse and the other survive?"*

Watch For

Have students turn and talk: What design choices could make one building survive and another fail?

Slide 4: LEVER Framework

2 min

Say: *"Our model has one thing from NATURE and one thing from ENGINEERING. The question is: can engineering beat nature?"*

Watch For

Preview components. Explain that this model is special because one external is a natural hazard and the other is a human solution.

Slide 5: Activity 1: Components

8 min

Say: *"We cannot control earthquakes — but we CAN control how we build. That is the key to this whole model."*

Watch For

Guide sorting. Earthquake Magnitude is external (nature). Building Flexibility is external (engineering choice). Three internals are results.

Slide 6: Activity 2: Connections

10 min

Say: *"Building Flexibility has a NEGATIVE relationship with Structural Damage. What does that mean? More flexibility means LESS damage!"*

Watch For

Help students draw all five relationships. Emphasize the two negative arrows — these are where engineering saves lives.

Slide 7: Activity 3: Simulation

12 min

Say: *"Now let us test it. Same earthquake, two different buildings. Watch what happens to Structural Damage and Occupant Safety."*

Watch For

Run the rigid vs. flexible comparison. Students record the dramatic difference in damage and safety outcomes.

Slide 8: Discoveries

5 min

Say: *"Engineers cannot stop earthquakes, but they CAN stop buildings from killing people. The answer is designing buildings that MOVE instead of FIGHT."*

Watch For

Show examples of base isolation and tuned mass dampers. Connect each engineering solution back to the model's flexibility component.

Slide 9: STEM Challenge

12 min

Say: *"Your mission: build the tallest tower that survives the earthquake test. But here is the twist — it also has to be tall enough to be useful!"*

Watch For

Distribute materials (straws, pipe cleaners, marshmallows, tape, clay). Groups build, test on shaking tables, iterate, and retest.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Earthquake Magnitude, Building Flexibility
- **INTERNAL:** Shaking Force, Structural Damage, Occupant Safety

Relationships

- **Earthquake Magnitude to Shaking Force:** POSITIVE (+) — Stronger earthquakes release more energy, creating stronger seismic waves that produce more violent shaking forces on buildings. A magnitude 7 earthquake produces about 32 times more energy than a magnitude 6, creating proportionally more shaking force.
- **Building Flexibility to Structural Damage:** NEGATIVE (-) — More flexible buildings experience LESS structural damage because they absorb and distribute shaking energy by swaying, bending, and flexing instead of cracking and collapsing. Flexibility allows the building to move WITH the earthquake rather than fighting against it.
- **Shaking Force to Structural Damage:** POSITIVE (+) — Stronger shaking forces cause more structural damage — more cracks, more broken connections, more collapsed sections. When shaking force exceeds what the structure can handle, failure is catastrophic and rapid.
- **Structural Damage to Occupant Safety:** NEGATIVE (-) — More structural damage means LESS safety for the people inside. When walls crack, floors collapse, and debris falls, occupants are at extreme risk. The goal of earthquake engineering is to keep structural damage low enough that the building protects everyone inside.
- **Building Flexibility to Occupant Safety:** POSITIVE (+) — More flexible buildings directly improve occupant safety because they maintain structural integrity during shaking. A building that sways but does not break keeps its occupants safe even in strong earthquakes.

Key Discoveries

- Stronger earthquakes create stronger shaking forces that cause more damage to buildings and endanger more people
- Flexible buildings survive earthquakes much better than rigid ones because they absorb and distribute shaking energy instead of fighting it
- Building flexibility REDUCES structural damage (negative relationship) because the building moves WITH the earthquake instead of against it
- Occupant safety depends directly on structural damage — the less damage, the safer the people inside, which is why engineering design saves lives

Multiple Choice Post-Assessment Answer Key

1. During an earthquake, a flexible building sways back and forth but does not collapse, while a rigid building next to it cracks and crumbles. Why does the flexible building survive?

Correct: B) The flexible building moves WITH the earthquake, absorbing and distributing the shaking energy instead of fighting against it

Correct! Flexible buildings survive because they sway, bend, and absorb shaking energy. Like a living tree bending in a storm instead of snapping, flexible buildings move with the earthquake rather than resisting it.

2. In the model, Earthquake Magnitude and Building Flexibility are external components. Why?

Correct: B) Because they are the starting conditions, one natural and one engineered, that determine what happens to the internal components

Right! Earthquake Magnitude is a natural condition we cannot control, and Building Flexibility is an engineering design choice. Together, these external inputs determine the Shaking Force, Structural Damage, and Occupant Safety inside the system.

3. The model shows that Building Flexibility has a NEGATIVE relationship with Structural Damage. What does this mean for earthquake engineering?

Correct: C) When Building Flexibility increases, Structural Damage decreases because the building absorbs shaking energy instead of cracking

Exactly! A negative relationship means when one goes up, the other goes down. More flexibility means less damage because the building can bend and sway to absorb the earthquake's energy without breaking apart.

4. A student compares two scenarios: a magnitude 90% earthquake hitting a building with 20% flexibility versus the same earthquake hitting a building with 80% flexibility. What difference would the model show?

Correct: B) The rigid building (20% flexibility) would have much more Structural Damage and much lower Occupant Safety than the flexible building (80% flexibility)

Right! Even though the earthquake is the same, the flexible building absorbs the shaking energy by bending and swaying. The rigid building fights the force, leading to cracks, collapses, and much greater danger to people inside.

5. The model shows that Structural Damage has a NEGATIVE relationship with Occupant Safety. Why is this the most important relationship for earthquake engineers?

Correct: B) Because when Structural Damage increases, Occupant Safety decreases, meaning cracking walls and falling debris put people in danger

That is right! This is the key relationship because it directly connects building performance to human lives. When walls crack, floors collapse, and debris falls, people inside are at extreme risk. Engineers design for flexibility to keep damage low and safety high.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The strongest building is the safest during an earthquake"	STRENGTH alone does not make a building earthquake-safe — FLEXIBILITY does. A building can be extremely strong but also extremely rigid, meaning it resists shaking forces until those forces exceed its strength, at which point it fails catastrophically. A flexible building may appear weaker but absorbs shaking energy by moving, preventing the catastrophic failure that kills people.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Earthquakes directly kill people"	Earthquakes themselves almost never directly kill people — BUILDINGS do. Falling debris, collapsing walls, and pancaking floors are responsible for the vast majority of earthquake deaths. In an open field during an earthquake, you would feel the shaking but be completely safe. This is why earthquake engineering is so important — better buildings mean fewer deaths, even in strong earthquakes.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We should just avoid building in earthquake zones"	About 1.5 billion people live in earthquake zones worldwide. Major cities like Tokyo, Los Angeles, San Francisco, Mexico City, and Istanbul are all in high-risk zones. We cannot simply abandon these areas — instead, we use engineering to make buildings safe. Japan proves this is possible: they experience thousands of earthquakes per year but their buildings are so well-engineered that most earthquakes cause no damage at all.